Financial Services -- Cortex Code HOL

Powered by Cortex Code

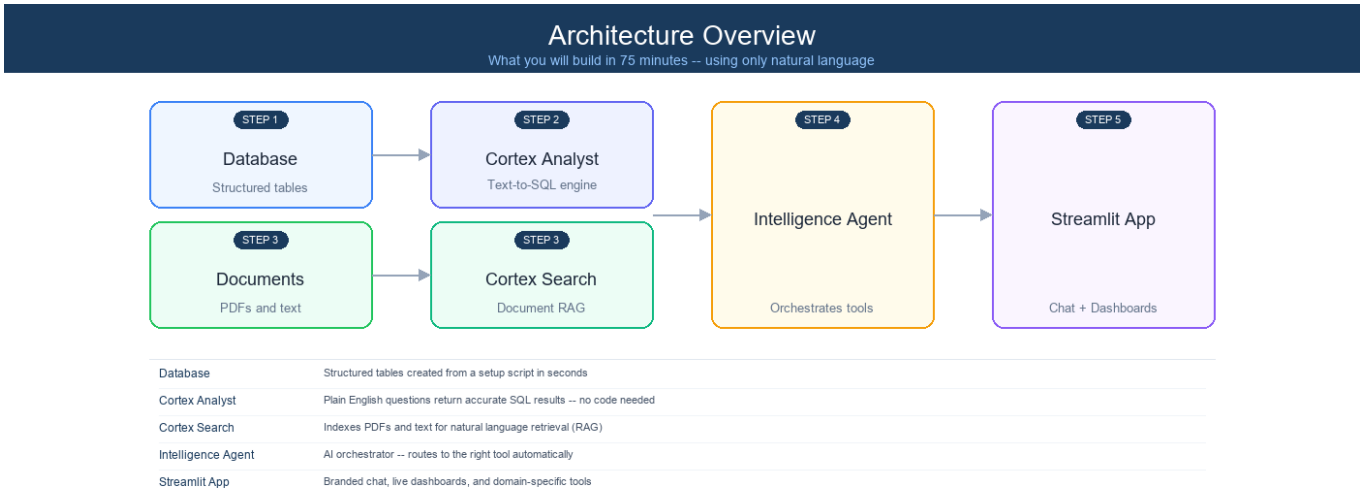
About This Lab

In this hands-on lab, you will build a fully functional AI-powered Portfolio Intelligence using **only natural language prompts** in Snowflake's AI-powered development platform, **Cortex Code**. No SQL writing, no YAML editing, no manual UI configuration -- you'll talk to Cortex Code and it will build everything for you.

What you'll build:

1. **Portfolio Database** -- Clients, holdings, risk scores, and compliance alerts
2. **Cortex Analyst** -- Ask questions in plain English, get SQL results instantly
3. **Cortex Search** -- Natural language search across risk management documents
4. **Intelligence Agent** -- AI orchestrator that routes queries to the right tool automatically
5. **Streamlit App** -- Chat interface, live dashboards, and portfolio rebalancing

Time: 75 minutes hands-on **Prerequisites:** A laptop with a modern browser. No coding experience required. No files to download -- everything comes directly from GitHub.



Database	Structured tables created from a setup script in seconds
Cortex Analyst	Plain English questions return accurate SQL results -- no code needed
Cortex Search	Indexes PDFs and text for natural language retrieval (RAG)
Intelligence Agent	AI orchestrator -- routes to the right tool automatically
Streamlit App	Branded chat, live dashboards, and domain-specific tools

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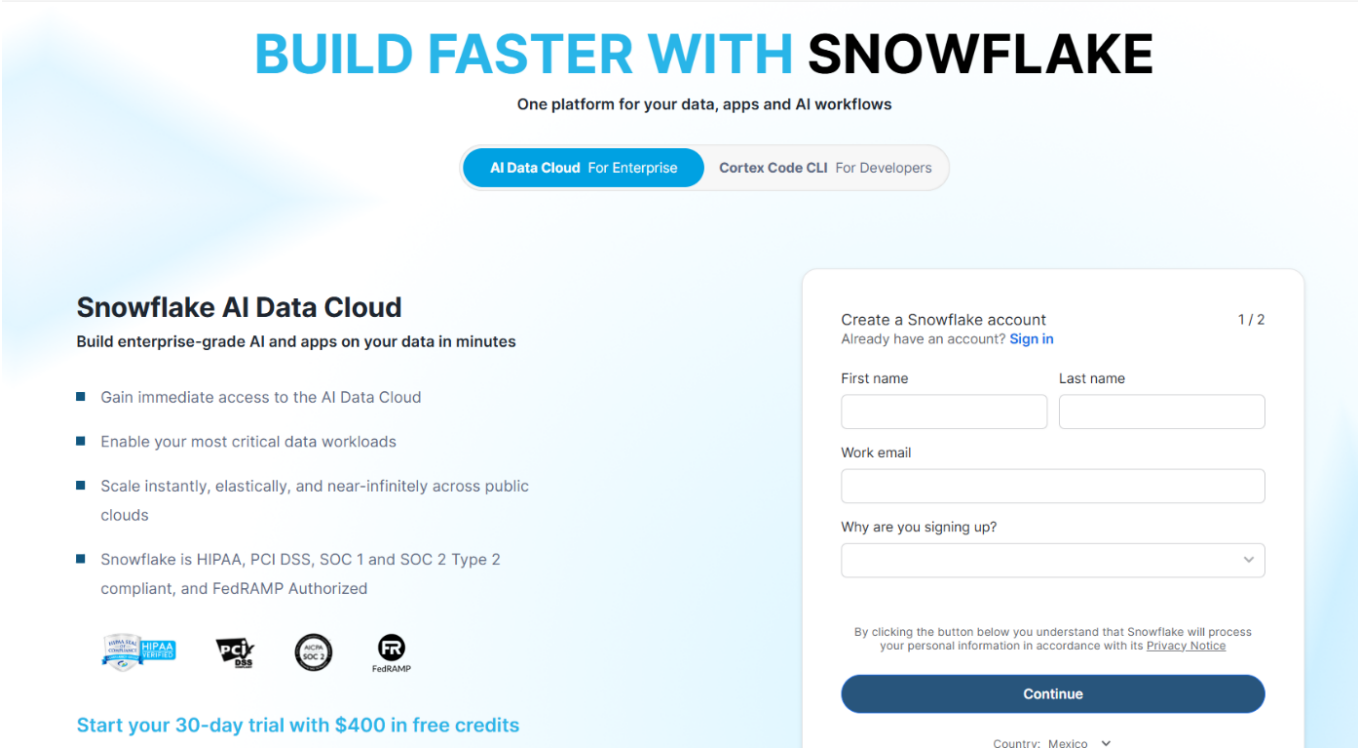
Step 0: Get Started (8 minutes)

0.1 Create and Log Into Your Snowflake Trial Account

A shared URL will be provided to sign up for a Snowflake trial account.

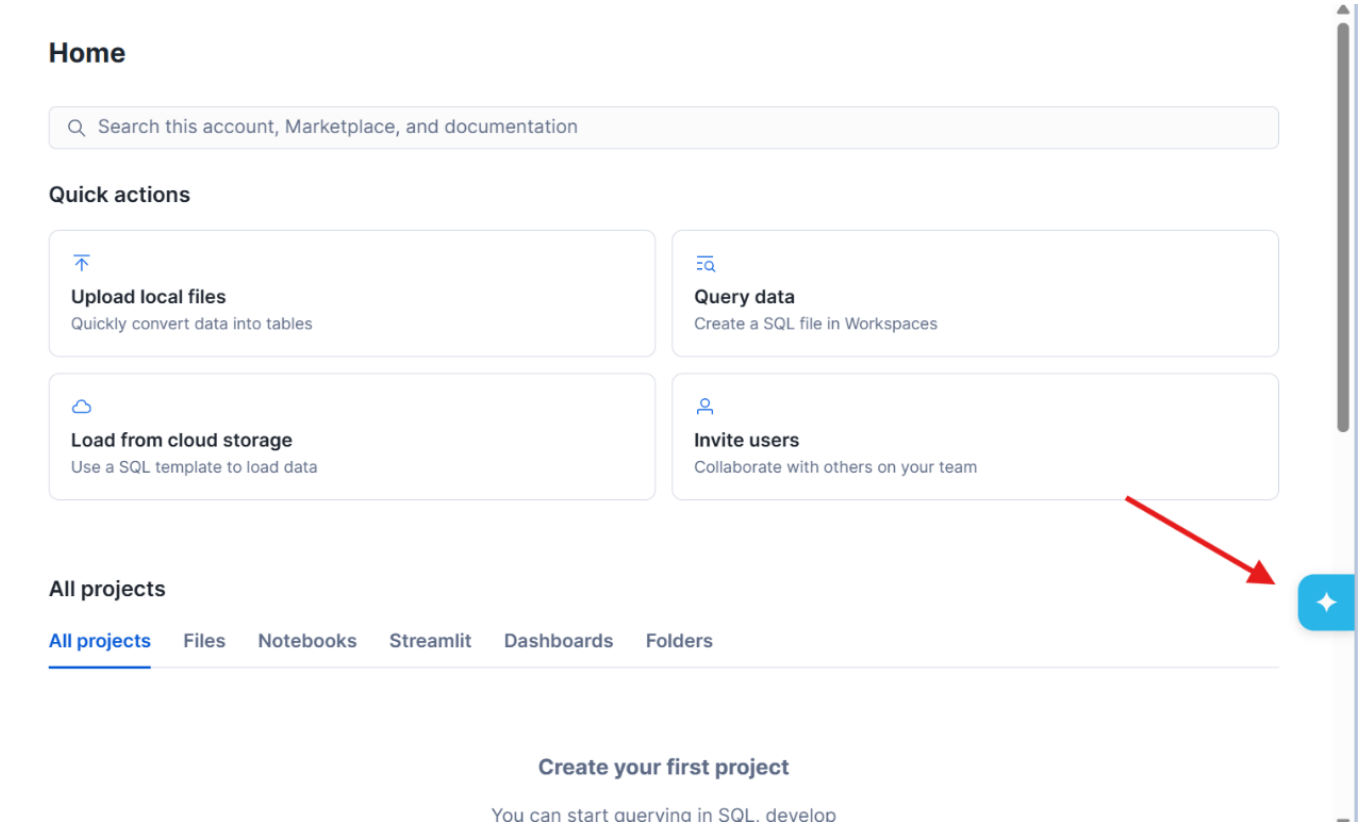
1. Navigate to the **signup URL** provided by your instructor
2. Ensure **"AI Data Cloud for Enterprise"** is selected (see screenshot below)
3. Select **AWS** as the cloud provider and **US East (Virginia)** as the region
4. Fill out the registration form with your details
5. Click **Continue** and complete the signup process
6. Check your email for a confirmation from Snowflake
7. Follow the instructions in the email to activate your account and set your password

8. Log in to your new Snowflake trial account — you should land on the **Snowsight** home page



0.2 Open Cortex Code

1. Look for the **Cortex Code icon** in the **lower-right corner** of Snowsight (see screenshot below)
2. Click it -- the Cortex Code panel will open on the right side of the screen
3. You should see a chat input box at the bottom of the panel



> **What is Cortex Code?** Cortex Code is Snowflake's AI-powered development platform. You can ask it to write SQL, create objects, build applications, and more -- all through natural language conversation. Think of it as your AI pair programmer that understands Snowflake natively.

0.3 Connect to the Lab's GitHub Repository

Instead of downloading any files, you'll connect Snowflake directly to the public GitHub repository that contains all the lab materials. Snowflake can read and execute files straight from GitHub -- no ZIP files, no uploads.

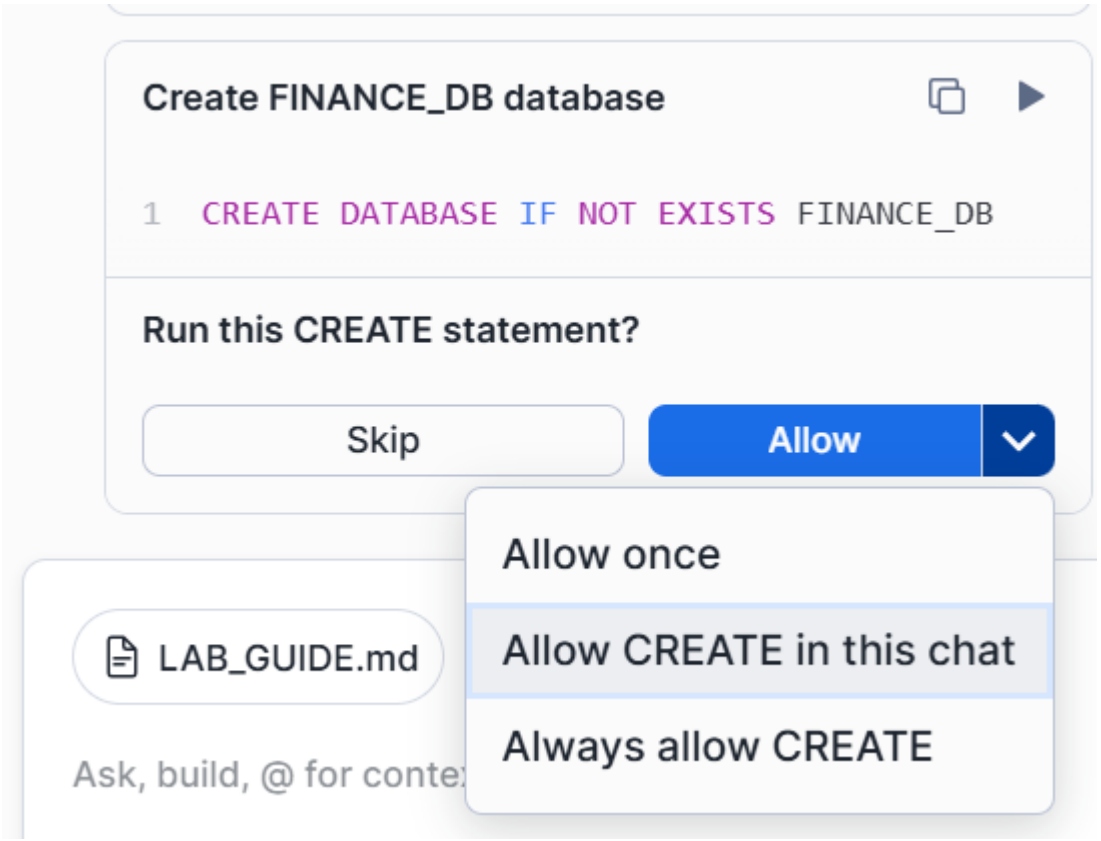
Copy and paste this prompt into Cortex Code:

```
Please connect my Snowflake account to a public GitHub repository so I can
use it as a file source during this lab.

1. Create a database called HOL_UTILS with schema PUBLIC if it doesn't exist
2. Create an API integration called HOL_GITHUB_API for GitHub HTTPS access,
   allowing the prefix https://github.com/jfromkamel/
3. Create a Git repository object called SNOWFLAKE_AI_HOL_REPO in the
   HOL_UTILS.PUBLIC schema pointing to:
   https://github.com/jfromkamel/snowflake-ai-hol
4. Fetch the latest contents from the repository
5. Confirm everything worked by listing the top-level files and folders
   available in the main branch
```

> **What's happening:** Cortex Code is creating a live connection between your Snowflake account and the lab's GitHub repository. All scripts, models, and documents will be accessible directly from GitHub -- no manual file handling needed.

> **Important -- "Allow in this chat" (or "Allow in this session") permissions:** When Cortex Code tries to run a CREATE statement, you'll see a permission prompt asking whether to allow it. Click the **Allow** dropdown arrow and select **"Allow CREATE in this chat"** (or **"Allow CREATE in this session"**). Do the same for any subsequent permission prompts you encounter (e.g., ALTER, INSERT, etc.). This prevents you from having to manually approve every single statement for the rest of the lab.



Expected output: In the Cortex Code response, you should see a folder listing including `healthcare/`, `financial-services/`, `industrial/`, and `README.md`.

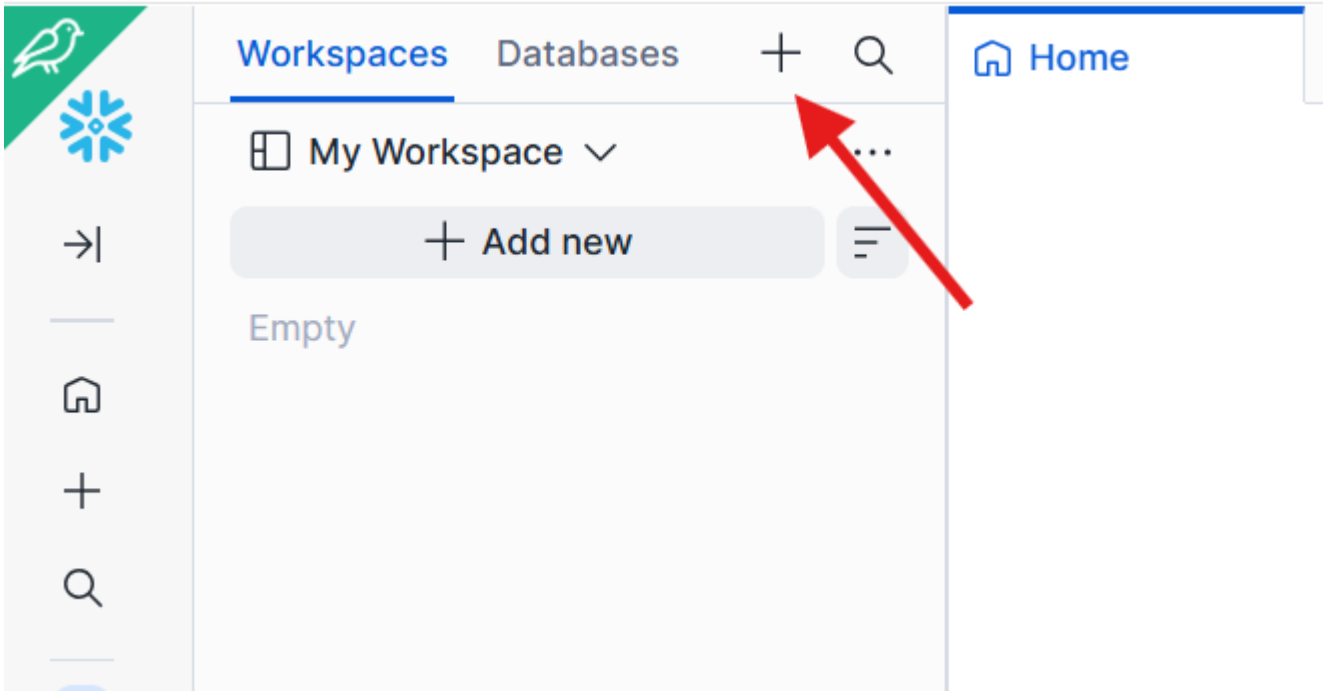
> **Note:** This is the only prompt in the lab that requires ACCOUNTADMIN. Cortex Code will handle the role switching automatically.

> **Troubleshooting:** If Cortex Code can't handle the Git setup, see **Appendix A: Fallbacks** at the end of this guide.

0.4 Browse Lab Files in a Workspace

Now that Snowflake is connected to GitHub, create a visual workspace to browse the repo files:

1. In the left sidebar, go to **Projects > Workspaces**
2. Click the **+** next to Databases and select **Git workspace**
3. Paste: `https://github.com/jfromkamel/snowflake-ai-hol`
4. Select `HOL_GITHUB_API` as the API Integration
5. Choose **Public repository** for authentication
6. Click **Create**



Expected output: The repo files (`healthcare/`, `financial-services/`, `README.md`) appear in the left-hand file explorer. You can browse any file directly from Snowsight.

> **Why this matters:** Your Snowflake account now has a live, bidirectional connection to GitHub. You can pull the latest changes, create branches, commit edits, push updates back to the remote repo, and resolve merge conflicts -- all directly from Snowsight. This means developers get full Git workflows (branching, pull/push, conflict resolution) natively inside Snowflake without switching tools.

> **Tip:** Feel free to browse the files freely in the workspace file explorer -- you'll find setup scripts, PDF documents, and prompt libraries for each industry track.

Step 1: Build the Database and Load Data (10 minutes)

In this step, you'll create a financial services database with clients, portfolios, holdings, risk scores, and compliance alerts -- all by asking Cortex Code to run a single script directly from GitHub.

1.1 Execute the Setup Script

Copy and paste this prompt into Cortex Code:

```
Please execute the setup script from the GitHub repository we connected.
Run this file to create the financial services database, warehouse, all
tables, internal stages, and load all sample data:

@HOL_UTILS.PUBLIC.SNOWFLAKE_AI_HOL_REPO/branches/main/financial-services/scripts/setup.sql

Use EXECUTE IMMEDIATE FROM to run it. After execution, set
FINSERV_HOL_WH as the active warehouse for the rest of our session
and show me a summary table of every table created with its schema and
row count so I can confirm everything loaded correctly.
```

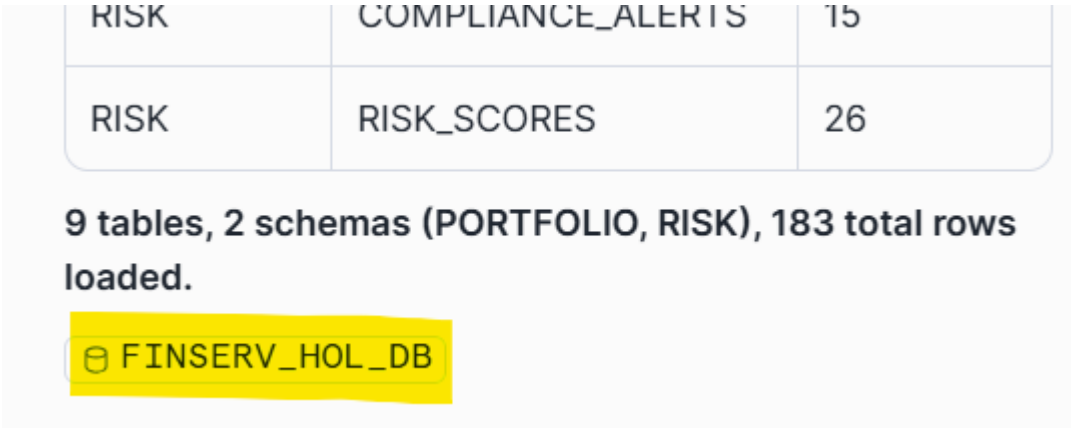
```
Include a pill at the end of the response that redirects to the created database.
```

> **What's happening:** The script creates: - Database: `FINSERV_HOL_DB` with schemas `PORTFOLIO` and `RISK` - Warehouse: `FINSERV_HOL_WH` - 9 tables with realistic financial data (clients, accounts, holdings, transactions, risk scores, compliance alerts, benchmarks) - Internal stages for files you'll use later

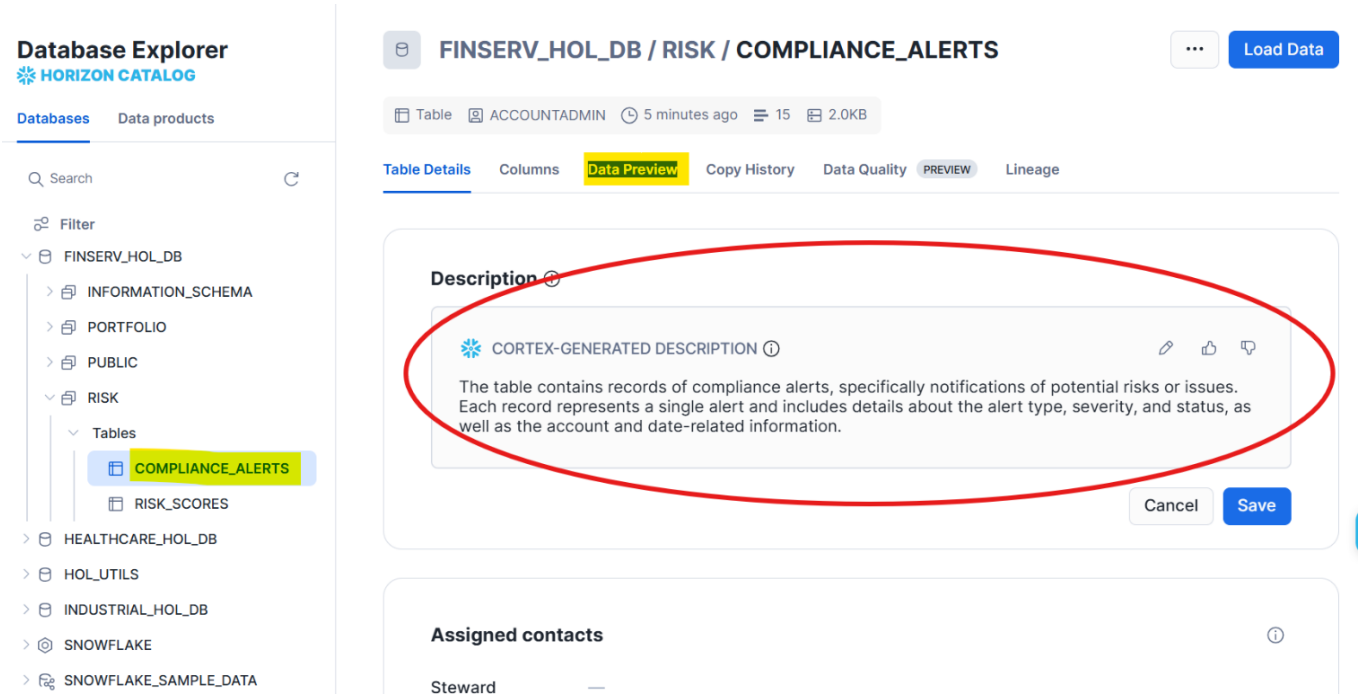
> **Congratulations!** You just built an entire database without writing a single line of SQL.

1.2 Explore Your New Database

Take a moment to see what was created. In the Cortex Code response, you'll see a hyperlink to **FINSERV_HOL_DB** — click it to open the database directly in the Database Explorer. From there, expand **RISK > Tables** and click on any table (e.g., **COMPLIANCE_ALERTS**) to see its details, columns, and a Cortex-generated description that automatically explains what the table contains.



> **Tip:** If Cortex Code didn't provide a hyperlink to the database in the response, you can navigate to it manually through the **Database Explorer** menu in the left sidebar. From there, expand **FINSERV_HOL_DB > RISK > COMPLIANCE_ALERTS**. See **Appendix A: Fallbacks** at the end of this guide for a screenshot.



> **Why this matters:** The Database Explorer gives you instant visibility into every object Cortex Code created -- table schemas, row counts, column types, and AI-generated descriptions -- without writing a single query.

Step 2: Create the Semantic Layer (12 minutes)

Cortex Analyst is Snowflake's text-to-SQL engine. Define your business metrics once -- then anyone can query portfolio data in plain English.

2.1 Create a Cortex Analyst Semantic View for Portfolio Data

Copy and paste this prompt into Cortex Code:

```
Create a Cortex Analyst semantic view for the portfolio data in
FINSERV_HOL_DB.PORTFOLIO.

Include these tables: ADVISORS, CLIENTS, ACCOUNTS, SECURITIES, HOLDINGS,
TRANSACTIONS, and BENCHMARKS.

Business intelligence rules:
- Flag "concentrated position" when POSITION_PCT > 15
- Flag "high risk" when RISK_SCORE > 0.8 (from FINSERV_HOL_DB.RISK.RISK_SCORES)

Make sure it can answer questions like:
1. "What is the total AUM by advisor?"
2. "Which accounts have concentrated positions above 15%?"
3. "Show me the top 10 holdings by unrealized P&L"
4. "What is the YTD performance vs S&P 500 benchmark?"

Name it PORTFOLIO_ANALYTICS and store it in FINSERV_HOL_DB.PORTFOLIO.
Register it directly as a semantic view without saving any intermediate
files to a stage.
```

2.2 Create a Cortex Analyst Semantic View for Risk Analytics

Copy and paste this prompt into Cortex Code:

```
Create a Cortex Analyst semantic view for market risk and compliance
analytics.

Include these tables:
- FINSERV_HOL_DB.RISK.RISK_SCORES
- FINSERV_HOL_DB.RISK.COMPLIANCE_ALERTS
- FINSERV_HOL_DB.PORTFOLIO.ACCOUNTS
- FINSERV_HOL_DB.PORTFOLIO.CLIENTS

Join RISK_SCORES to ACCOUNTS on ACCOUNT_ID, COMPLIANCE_ALERTS to
ACCOUNTS on ACCOUNT_ID, and ACCOUNTS to CLIENTS on CLIENT_ID.

Business intelligence rules:
- Flag "high risk" when RISK_SCORE > 0.8
- Flag "critical alert" when SEVERITY = 'Critical'
- Flag "open alert" when STATUS = 'Open'
- Calculate total Value at Risk (VaR) at 95% confidence from VAR_95

Make sure it can answer questions like:
1. "What is the total Value at Risk across all high-risk accounts?"
2. "Show me all open compliance alerts by severity"
3. "Which clients have the highest portfolio risk scores?"
4. "What is the average Sharpe ratio by account type?"

Name it MARKET_RISK and store it in FINSERV_HOL_DB.RISK.
Register it directly as a semantic view without saving any intermediate
files to a stage.
```

2.3 Explore Your Semantic Model in Snowsight

Step out of Cortex Code for a moment -- let's see what you built in the Snowsight UI.

Navigate to the Analyst:

1. In the left navigation, hover over the **Cortex** icon (or **AI/ML** section)
2. Click **Analyst**
3. In the **Database** dropdown at the top, select **FINSERV_HOL_DB**, then select the **PORTFOLIO** schema
4. Find your **PORTFOLIO_ANALYTICS** semantic view in the list and click it to open it

Ask a question:

1. Click the **Playground** tab on the right-hand side to open the chat interface
2. In the chat box, type: *"What is the total AUM by advisor region?"*
3. Cortex Analyst generates SQL, runs it, and returns results -- all in one step
4. Click **Add to Verified Queries** below the response to save this as a verified query. Verified queries act as trusted reference answers that improve the model's accuracy over time.

Explore your semantic view:

1. Click the **Suggestions** tab and then click **Start Learning** to let the model analyze your data. While it learns, explore the other tabs:
2. **Custom Instructions** -- business rules and context you provided (e.g., "concentrated position = POSITION_PCT > 15")
3. **Logical Tables** -- the tables included in this model and their columns
4. **Relationships** -- how tables are joined together
5. **Verified Queries** -- saved question/SQL pairs (including the one you just added) that serve as reference examples
6. **Monitor** -- a history of every question asked, the SQL generated, whether it succeeded, and user feedback

> **Note:** The Analyst can answer *any* natural language question about the included tables -- not just verified ones. Verified queries simply improve accuracy by giving the model reference examples of correct SQL for common questions.

Ask a follow-up:

1. Go back to the **Playground** tab and ask: *"Which clients have the highest unrealized gains?"*
2. Notice the conversation maintains context -- this is a full dialogue, not isolated one-off queries.
3. Return to the **Suggestions** tab -- check if the model has generated any suggestions for improving your semantic view's coverage.

> **Key takeaway:** Any advisor, analyst, or executive can now query portfolio data in plain English. No SQL required -- and every answer is grounded in the same business logic you just defined.

Step 3: Set Up Document Search (8 minutes)

So far, your agent can answer any question about risk management policies, rebalancing guidelines, concentration limits, and compliance rules that lives in structured tables. But a lot of critical knowledge lives somewhere else entirely -- in documents.

When a client asks "What is our policy on concentrated positions?", the answer isn't in a database row -- it's in your Risk Management Framework document. That's the gap Cortex Search fills.

In this step, you'll turn a PDF document into a fully searchable knowledge base. Cortex Search automatically reads the document, breaks it into meaningful sections, and builds a semantic index so the agent can retrieve the right passage for any question -- even when the exact words don't match.

This is what's commonly called **RAG (Retrieval Augmented Generation)** -- but you won't need to configure any of that. Cortex Code handles it in one prompt.

3.1 Bring the PDF into Snowflake

> **Before continuing:** Click the **Snowflake logo** in the upper-left corner to return to the homepage. If the Cortex Code chat panel is closed, expand it from the lower-right corner.

Copy and paste this prompt into Cortex Code:

```
Bring the following PDF from our GitHub repository into
FINSERV_HOL_DB.PORTFOLIO so we can make it searchable:
```

```
@HOL_UTILS.PUBLIC.SNOWFLAKE_AI_HOL_REPO/branches/main/financial-services/pdfs/Risk
Management Framework.pdf
```

3.2 Create the Search Service

Copy and paste this prompt into Cortex Code:

```
Create a Cortex Search service called RISK_DOCS_INFO in
FINSERV_HOL_DB.PORTFOLIO that makes the risk management PDF
searchable with natural language.

To do this:
- Temporarily scale up FINSERV_HOL_WH for processing power
- Read and parse the PDF content
- Break it into overlapping chunks for better search accuracy
- Index everything so it can be queried in plain English
- Scale the warehouse back down to SMALL when done
```

3.3 Preview the Document and Explore the Search Service

First, preview the source PDF:

Before testing search, let's confirm the document loaded correctly and give you a chance to review it. Paste this prompt into Cortex Code:

```
Generate a presigned URL so I can preview the file "Risk Management Framework.pdf" in
FINSERV_HOL_DB.PORTFOLIO.PDF_STAGE -- use an expiry of 3600 seconds.
```

Cortex Code will return a URL. Open it in a new browser tab to read the source document your search service is built on.

> **Why this matters:** Seeing the source document helps you understand why the search returns what it returns. You're not searching a black box -- the content is fully visible and auditable.

Navigate to the Search Playground:

1. In the left navigation, hover over the **Cortex** icon (or **AI/ML** section)
2. Click **Cortex Search**
3. In the **Database** dropdown, select **FINSERV_HOL_DB** (might already be pre-selected)
4. Find **RISK_DOCS_INFO** in the list and click on it to open it
5. You'll land on the service detail page -- click the **Query** tab (or **Playground**) to open the search interface

Run a search: 6. In the search box, type: *"portfolio rebalancing triggers"* 7. Review the results -- notice it returns the most relevant passages from the document, not just keyword matches 8. Click into a result to see the full chunk of text it retrieved

Try a semantic search (the real power): 9. Now type: *"maximum position concentration limits"* 10. Try rephrasing the same concept in different words -- notice the results stay relevant even when your wording doesn't exactly match the document

> **What you're seeing:** This is hybrid search -- combining keyword matching with semantic (meaning-based) matching. It finds the right passage even when someone asks a question differently than the document is written.

> **Key takeaway:** Your agent can now answer compliance questions like "What are our rebalancing thresholds?" -- grounded in your actual risk framework, not hallucinated.

Step 4: Build the Intelligence Agent (10 minutes)

4.1 Create the Agent

Copy and paste this prompt into Cortex Code:

```
Create a Snowflake Intelligence Agent named FINSERV_AGENT in
FINSERV_HOL_DB.PORTFOLIO.

Give it three tools:

1. A data analysis tool named PORTFOLIO_DATA connected to the
  PORTFOLIO_ANALYTICS semantic model in FINSERV_HOL_DB.PORTFOLIO.
  It answers questions about clients, accounts, holdings,
  transactions, and performance. Use FINSERV_HOL_WH.

2. A data analysis tool named RISK_DATA connected to the
  MARKET_RISK semantic model in FINSERV_HOL_DB.PORTFOLIO.
  It analyzes risk scores, compliance alerts, and portfolio
  risk metrics. Use FINSERV_HOL_WH.

3. A document search tool named RISK_DOCS connected to the
  RISK_DOCS_INFO search service in FINSERV_HOL_DB.PORTFOLIO.
  It searches risk management documentation for policies
  and compliance rules.
```

4.2 Explore the Two UIs

There are two separate places in Snowsight where your agent lives -- and they serve completely different audiences.

View 1: Cortex > Agents (the admin view)

This is where developers and data engineers build and manage agents. Think of it as the control panel.

1. In the left navigation, hover over the **Cortex** icon (or **AI/ML** section)
2. Click **Agents**
3. Find **FINSERV_AGENT** and click on it
4. On the left-hand side, you'll see all 3 tools listed:
5. PORTFOLIO_DATA (Cortex Analyst -- portfolio data)
6. RISK_DATA (Cortex Analyst -- risk and compliance)
7. RISK_DOCS (Cortex Search -- documentation)
8. Ask a couple of questions in the chat panel, for example: *"Which accounts have concentrated positions above 15%?"* and *"What is the rebalancing policy for high-risk portfolios?"*
9. Open the **Monitor** tab and click on a specific request to see all monitoring metrics -- latency, tokens used, tool calls made, and the full response trace

> **Who uses this view:** Admins and data engineers. This is where you build, update, and govern agents. Business users never need to come here.

View 2: Snowflake Intelligence (the end-user experience)

This is what your business users actually see -- a polished, conversational interface built on top of the same agent.

1. Click **Preview in Snowflake Intelligence** in the upper-right corner to switch from the admin view to the end-user experience

What's different here compared to Cortex > Agents: - Responses automatically render as **charts or tables** depending on the question type -- no configuration needed - Citations appear below answers, linking back to the exact data source or document passage - Conversations are **threaded** -- users can continue a conversation, branch off, or start fresh - Users can **upload files** (CSV, PDF, XLSX) directly in the chat to add context to their questions - There is an **Extended Thinking** toggle for complex, multi-step analytical questions

> **Why this matters:** The agent you built in Cortex Code is the engine. Snowflake Intelligence is the car. Business users interact with the car -- they never need to know what's under the hood.

4.3 Test the Agent in Snowflake Intelligence

Try these questions to see the three types of responses:

Question 1 -- Structured data (table response):

> Which accounts have concentrated positions above 15%?

Question 2 -- Charting (visual response):

> Plot the total assets under management by advisor region as a bar chart.

Question 3 -- Unstructured data (document search):

> What does our risk management framework say about rebalancing trigger thresholds?

Step 5: Build a Streamlit App (25 minutes)

You've already built something powerful: an agent accessible through Snowflake Intelligence with conversational AI, auto-generated charts, threaded conversations, and citations. So what does a Streamlit app add on top of that?

Think about an advisor who needs a client review dashboard showing portfolio performance, risk scores, and rebalancing recommendations in a structured, branded layout. That's a different kind of experience -- a purpose-built application with a fixed layout, specific operational views, and embedded computation like the optimization engine you'll build in this step.

The agent you've created is actually surfaceable in three different ways, and they complement each other:

	Snowflake Intelligence	Streamlit in Snowflake	Cortex Agents REST API
Best for	Open-ended exploration and self-service analytics	Structured operational apps with fixed layouts and custom tools	Embedding the agent anywhere outside Snowflake
Experience	Conversational with auto-charting, threading, and citations	Multi-page app with dashboards, forms, and Python computation	Any interface you build -- chat widgets, portals, integrations
Customization	Agent behavior and instructions	Full control over UI, layout, branding, and logic	Full control -- wire the agent into Slack, Teams, web apps, or mobile
Computation	Answers and analysis from your data	Python on top of live data -- optimization, ML, file processing	Depends on the client you build

> **The Cortex Agents REST API** is the same endpoint this Streamlit app will call. That means the agent you built today can also power a Slack bot, a Microsoft Teams integration, an external web portal, or a mobile app -- with no changes to the agent itself.

In this step, you'll build a 3-page app: 1. **Chat** -- the same agent, now inside your own branded interface 2. **Dashboard** -- fixed operational views of AUM by advisor, open compliance alerts, and concentration risk flags 3. **Optimization** -- a portfolio rebalancing optimizer built in Python that runs directly on your live data

> **The big picture:** Snowflake Intelligence is a great out-of-the-box experience. Streamlit is how you build a product around the same intelligence layer -- and the REST API is how you take it everywhere else.

5.1 Create the Chat Page (8 minutes)

> **Before continuing:** Switch back to your original browser tab (Snowflake Intelligence opened in a new tab). Navigate to **Projects > Workspaces** and open your workspace. Make sure Cortex Code is open in the lower-right corner.

Copy and paste this prompt into Cortex Code:

```
Build a single-page Streamlit app in Snowflake called FINSERV_ASSISTANT_APP
in FINSERV_HOL_DB.PORTFOLIO using FINSERV_HOL_WH.

Use my workspace to create the Streamlit files directly for collaboration.
Use streamlit==1.52.2 for chat features.

Use st.tabs() to create a horizontal tab bar at the top of the page
with three tabs: "Chat", "Dashboard", and "Optimization". All three
tabs should be visible and switchable at the top -- no sidebar page
navigation.

Start with the Chat tab:
- A sidebar with the title "Portfolio Intelligence" and a brief description of the app
- A chat interface where users can have a conversation with FINSERV_AGENT
- Show the agent's responses in the chat as they arrive
- When the agent returns data, display it as a table below the response
- When the agent returns SQL, show it in a collapsible section
- Keep the conversation history visible throughout the session
```

```
Apply a premium financial services design throughout the app:
- Main content background: white (#FFFFFF) with light gray panels (#F8FAFC)
 and subtle card shadows -- clean and professional, not dark
- Sidebar: deep navy (#0F172A) with white text and a gold left border on the
 active page -- the sidebar is the only dark element
- Primary color: deep navy (#0F172A) for headings and key text; gold (#B45309)
 as a warm accent for section dividers, active states, and highlights
- Typography: sentence case throughout -- no all-caps; use semibold weight for
 headers, regular for body; confident and readable, not aggressive
- Data tables: white background, light gray dividers, numbers right-aligned;
 gains in green (#16A34A), losses in red (#DC2626); clean and data-dense
- KPI cards: white background with a gold top border, large bold number in
 dark navy (#0F172A), small gray label -- premium but not garish
- Charts: white background with deep navy as the primary series color and gold
 as the secondary -- consistent with the rest of the app
- Chat input area: white background matching the main content -- seamlessly
 part of the page, not a floating element
- Overall feel: a premium wealth management portal -- authoritative, polished,
 and trustworthy; think a private bank client dashboard, not a trading terminal
```

5.2 Open and Test the Chat Page

1. In the left sidebar, go to **Projects > Streamlit**, right-click **FINSERV_ASSISTANT_APP**, and select **Open in new tab**
2. Keep this tab open for the rest of the lab -- use it to view and refresh the app whenever changes are made. Use your other tab (with the workspace and Cortex Code) for prompting.

> **App not loading or showing an error?** Copy the full error message, go back to Cortex Code, and paste this prompt:
>
> ` My Streamlit app is showing the following error. Please read the app code, identify the cause, and fix it:
>
> [paste your error here] `
>
> Cortex Code will read the app files, diagnose the issue, and update the code. Refresh the app once it confirms the fix.
>
> **Visual design issues?** Cortex Code supports images -- take a screenshot of the problem, go back to Cortex Code, paste the screenshot directly into the chat, and use this prompt:
>
> `Here is a screenshot of my Streamlit app showing a visual issue. Please read the app code, identify what is causing it, and fix it.`
>
> Cortex Code will read the screenshot, diagnose the layout or styling problem, and update the code. Refresh the app once it confirms the fix.

1. Try: **"Which clients have the highest unrealized gains?"**

5.3 Add the Dashboard Tab (10 minutes)

Copy and paste this prompt into Cortex Code:

```
Fill in the Dashboard tab of FINSERV_ASSISTANT_APP with the following content.

This page should show live data from FINSERV_HOL_DB:

TOP ROW - four summary numbers:
- Total AUM (sum of all account balances)
- Active Clients (count from CLIENTS where STATUS = 'Active')
- Open Compliance Alerts (from RISK.COMPLIANCE_ALERTS where STATUS = 'Open')
- Average Portfolio Risk Score (from RISK.RISK_SCORES)

MIDDLE ROW - two charts side by side:
- A bar chart of AUM by advisor, showing each advisor's total
 assets under management
- A pie chart of holdings by asset class (Equity, Fixed Income,
```

```
 Alternative)

BOTTOM ROW - a compliance alerts table:
- All open alerts showing: Account Name, Client Name, Alert Type,
 Severity, Description, Days Open
- Sort by severity (Critical > High > Medium > Low)
- Include a button to download the table as a CSV

INTERACTIVITY:
- Add an advisor filter dropdown at the top that filters all charts and the table
- Add a severity multi-select filter for compliance alerts
- Make the bar chart bars clickable -- clicking an advisor filters the alerts table to
 show only that advisor's open alerts

Chart styling:
Use plotly for all charts. Color palette: deep navy (#0F172A) as the primary series color,
warm gold (#B45309) as the secondary. Use green (#16A34A) for gains/positive and red
(#DC2626) for losses/negative. All chart backgrounds should be white (#FFFFFF) with light
gray grid lines. No dark backgrounds on any chart.
Position the legend below each chart (orientation="h", y=-0.3) to prevent overlap with
chart titles. Add a top margin (t=50) on all charts so the title never crowds the plot
area.
```

5.4 Add the Optimization Tab (7 minutes)

Copy and paste this prompt into Cortex Code:

```
Fill in the Optimization tab of FINSERV_ASSISTANT_APP with the following content.

Title: "Portfolio Intelligence"
Subtitle: "Rebalance portfolios and forecast performance using
AI and machine learning on your live portfolio data."

At the top of the tab, add a mode selector using radio buttons:
- "Rebalancing Optimizer"
- "Performance Forecaster"

---

MODE 1: Rebalancing Optimizer

DATA EXPLORATION (shown immediately when this mode is selected):
- An advisor dropdown filter (from ADVISORS table, plus "All advisors")
- A risk tolerance multi-select (Conservative / Moderate / Aggressive)
- A plotly treemap showing current portfolio allocation by asset class
 across filtered accounts -- cells colored by concentration level
 (green within limits, amber approaching, red exceeding 15%)
 Updates in real-time as filters change.
- Below the treemap, a data table of holdings for the filtered accounts
 (Account, Security, Asset Class, Position %, Market Value, Status)

OPTIMIZATION CONTROLS (below the data exploration):
- A slider for Concentration Limit: 5% to 25% (default 15%)
- A steel blue "Run Optimizer" button

When clicked:
1. Pulls holdings from HOLDINGS filtered by the selected advisor and
  risk tolerance, finding positions where POSITION_PCT exceeds the limit
2. Uses linear programming (scipy) to generate sell/buy recommendations
  that bring each account back within the selected concentration limit
  while minimizing total notional trade value (fewest trades possible)
3. Respects client risk tolerance: Conservative accounts only receive
  fixed income buy suggestions, Aggressive accounts allow equity
```

```
Display results as:
- Three KPI cards: Accounts Needing Rebalancing, Total Notional Trade
 Value, Estimated Compliance Alerts Resolved
- A plotly treemap showing before/after allocation comparison
- A recommended sells table: Account, Security, Current %, Target %,
 Shares to Sell, Estimated Proceeds
- A recommended buys table: Security, Asset Class, Shares to Buy,
 Estimated Cost, Rationale
- A button to download recommendations as CSV

---

MODE 2: Performance Forecaster

DATA EXPLORATION (shown immediately when this mode is selected):
- An account dropdown (from ACCOUNTS table)
- A date range slider to explore historical data (last 90 days)
- A plotly line chart showing historical daily portfolio value from
 DAILY_MARKET_DATA for the selected account, updating as filters change.
 Use deep navy for portfolio value.
- Below the chart, a summary table of the filtered market data
 (Date, Portfolio Value, Daily Change %, Cumulative Return %)

FORECASTING CONTROLS (below the data exploration):
- Radio buttons for Forecast Horizon: 14 days / 30 days / 60 days
- A steel blue "Generate Forecast" button

When clicked:
1. Pull daily portfolio values from DAILY_MARKET_DATA for the selected
  account
2. Use SNOWFLAKE.ML.FORECAST to train a forecast model on the historical
  PORTFOLIO_VALUE and predict the next N days (based on the selected
  horizon). Run this as a SQL operation inside Snowflake.
3. Retrieve forecast values with upper and lower confidence bounds

Display results as:
- A headline KPI card: "Forecasted Portfolio Value" at end of horizon,
 with an arrow and color indicator (green if increasing, red if decreasing)
- A plotly line chart combining historical and forecast:
 - Historical portfolio value (solid navy line, #0F172A)
 - Forecasted value (dotted line with light gold confidence band)
 - X-axis: dates, Y-axis: dollar value
 - Legend below the chart, white background
- A risk outlook table: showing periods where the forecast lower bound
 drops below a key threshold (e.g., 5% below current value), with columns:
 Date, Forecasted Value, Lower Bound, Potential Drawdown %, Risk Level
- A button to download the forecast as CSV

---

Apply the same styling as the rest of the app throughout:
white background, deep navy sidebar (#0F172A), gold accent (#B45309),
plotly white chart backgrounds, sentence case, no all-caps.

Add scipy to the app's dependencies.
```

5.5 Test Your Complete App

1. **Chat** -- Ask "What is the total AUM by region?"
2. **Dashboard** -- Review compliance alerts and AUM distribution
3. **Optimization** -- Click "Run Optimization" and see rebalancing recommendations

> **Congratulations!** You've built a complete AI-powered financial services application -- all by talking to Cortex Code.

Conclusion: What You Built

In 90 minutes, using only natural language prompts, you built:

1. **A portfolio database** with 10 tables and 90 days of daily market data
2. **Two semantic models** enabling text-to-SQL analytics for any advisor or analyst
3. **A document search engine** (RAG) over your risk management framework
4. **An intelligent agent** that routes questions to the right data source automatically
5. **A 3-page Streamlit app** with chat, interactive dashboards, and ML-powered forecasting

Time Comparison: Cortex Code vs Traditional Development

Component	Traditional Approach	With Cortex Code
Database + sample data	2-3 days (schema design, ETL, testing)	~5 minutes
Semantic model	1-2 weeks (data modeling, business rules, testing)	~5 minutes
Document search (RAG)	1-2 weeks (parsing, chunking, embedding, indexing)	~3 minutes
AI agent with 3 tools	2-4 weeks (orchestration, routing, testing)	~5 minutes
Streamlit app (chat + dashboard + ML)	3-6 weeks (frontend, backend, ML pipeline)	~25 minutes
Total	**8-15 weeks**	**~45 minutes**

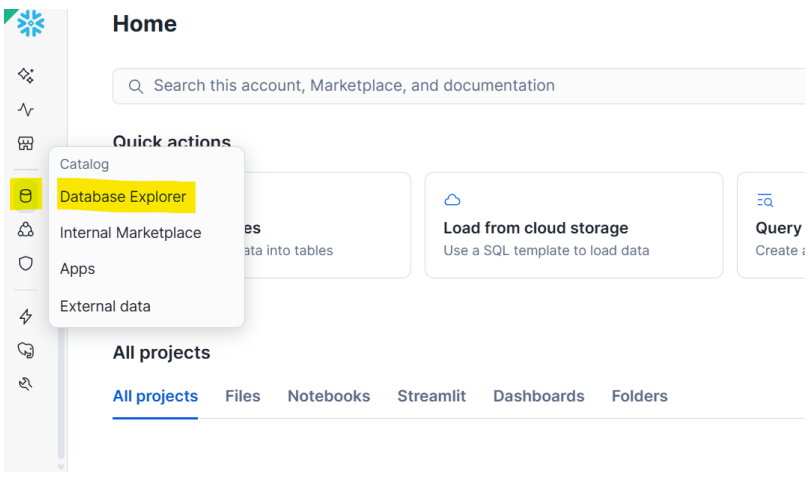
Key Takeaways

- **No code written manually** -- every artifact was generated through conversation
- **Production-grade output** -- semantic models, agents, and apps are real Snowflake objects you can share, govern, and scale
- **Three surfaces for one agent** -- Snowflake Intelligence (out-of-the-box), Streamlit (custom apps), REST API (embed anywhere)
- **ML built in** -- SNOWFLAKE.ML.FORECAST and scipy optimization running directly on live data, no external infrastructure

Appendix A: Fallbacks

Fallback for Step 1.2

If Cortex Code didn't provide a database hyperlink, use the **Database Explorer** menu in the left sidebar to navigate to your database:



Fallback SQL for Step 0.3

If Cortex Code can't handle the Git setup, run this SQL directly in a Snowsight worksheet:

```
USE ROLE ACCOUNTADMIN;
CREATE DATABASE IF NOT EXISTS HOL_UTILS;
CREATE SCHEMA IF NOT EXISTS HOL_UTILS.PUBLIC;
CREATE OR REPLACE API INTEGRATION HOL_GITHUB_API
 API_PROVIDER = git_https_api
 API_ALLOWED_PREFIXES = ('https://github.com/jfromkamel/')
 ENABLED = TRUE;
CREATE OR REPLACE GIT REPOSITORY HOL_UTILS.PUBLIC.SNOWFLAKE_AI_HOL_REPO
 API_INTEGRATION = HOL_GITHUB_API
 ORIGIN = 'https://github.com/jfromkamel/snowflake-ai-hol';
ALTER GIT REPOSITORY HOL_UTILS.PUBLIC.SNOWFLAKE_AI_HOL_REPO FETCH;
LS @HOL_UTILS.PUBLIC.SNOWFLAKE_AI_HOL_REPO/branches/main/;
```